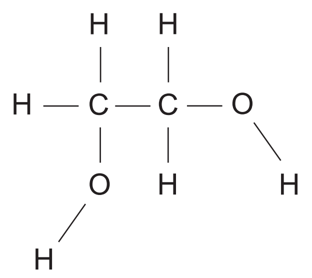
Examiner only

13. (a) In cold weather the wings of an aeroplane can become covered in ice. For safety reasons the wings must be de-iced.

The liquid ethane-1,2-diol, CH_2OHCH_2OH, is used, mixed with water, since this lowers the freezing temperature of water and causes the ice to melt.

(i) Use the diagram below to show the intermolecular forces that allow ethane-1,2-diol to dissolve in water. [3]



(ii) Suggest a reason why the addition of ethane-1,2-diol lowers the freezing temperature of water. [1]

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(b) Ethane-1,2-diol can be oxidised to ethanedioic acid, $(COOH)_2$.

(i) Suggest an oxidising agent suitable to carry out this reaction. [1]

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(ii) Write the equation for this reaction. Use [O] for the oxidising agent. [2]

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(iii) To ensure complete oxidation the reagents are refluxed.

Draw and label the apparatus as it is being used in this reaction. [2]

(iv) A sample of the reacting mixture was taken during the reflux process and a mass spectrum was produced. One of the peaks recorded was at m/z 58.

Suggest the identity of the **molecular ion** that gave this peak. [1]

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(v) The reaction can be used to prepare a sample of solid ethanedioic acid. This is generally hydrated as $(COOH)_2.xH_2O$ where x is an integer.

2.00 g of ethane-1,2-diol was oxidised and 3.94 g of hydrated ethanedioic acid was produced.

Calculate the relative molecular mass of hydrated ethanedioic acid and hence the value of x in its formula. [4]

x = ...

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(vi) I. Complete the equation for the reaction which occurs when ethanedioic acid is heated with excess methanol in acidic conditions.

Clearly show the structure of the organic product. [2]

$$\begin{array}{l} COOH \\ | \\ COOH \end{array} \quad + \quad 2CH_3OH \longrightarrow \qquad\qquad\qquad + \qquad\qquad$$

II. Name the type of functional group present in the organic product. [1]

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END OF PAPER